Advanced Image Synthesis

Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
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1. Graphics Pipeline

During this chapter, we will discuss the graphics pipeline, which is a sequence of steps that a computer graphics system uses to render a 3D scene onto a 2D screen. It starts from the initial creation of 3D models and ends with the final display of the rendered image. There are two types of graphics pipelines: the fixed-function pipeline and the programmable pipeline.

1.1. Fixed-Function Pipeline

The fixed-function pipeline is a traditional graphics pipeline that was widely used in older graphics hardware. It consists of a series of predefined stages that perform specific tasks. Each stage has a fixed function, and the data flows through these stages in a specific order. During the following subsections, we will discuss the stages of the fixed-function pipeline in detail.

1.1.1. Geometry

During this stage, the 3D models are created and defined. This includes defining the vertices, normals, texture coordinates, and other attributes of the 3D objects. In order to create the 3D models, the following primitives are used:

- `GL_POINTS`
- `GL_LINES`: A series of disconnected straight lines.
- `GL_LINE_STRIP`: A series of connected lines.
- `GL_LINE_LOOP`: A series of connected lines that form a loop.
- `GL_POLYGON`: A filled polygon defined by a series of vertices.
- `GL_TRIANGLES`: A series of disconnected triangles.
- `GL_TRIANGLE_STRIP`: A series of connected triangles. The first three vertices define the first triangle, and each subsequent vertex forms a new triangle with the previous two vertices.
- `GL_TRIANGLE_FAN`: A series of connected triangles that share a common central vertex. The first vertex is the center of the fan, and each subsequent vertex forms a new triangle with the center vertex and the previous vertex.
- `GL_QUADS`: A series of disconnected quadrilaterals.

- `GL_QUAD_STRIP`: A series of connected quadrilaterals. The first four vertices define the first quadrilateral, and each subsequent pair of vertices forms a new quadrilateral with the previous two vertices.

Even though all of these primitives are supported by OpenGL, it is recommended to use triangles for rendering, as they are the most efficient and widely supported primitive in modern graphics hardware.

In order to specify the vertices of the primitives, a process evolved from immediate mode to vertex arrays and eventually to vertex buffer objects (VBOs).

1. Immediate Mode: In this mode, vertices are specified one at a time. It is simple to use but inefficient, as it requires multiple function calls for each vertex. The geometry is then stored in the code.

2. Vertex Arrays: This mode allows you to specify an array of vertices and their attributes, which can be more efficient than immediate mode. The geometry is then stored in the RAM.

3. Vertex Buffer Objects (VBOs): This mode allows you to store vertex data in the GPU's memory, which can significantly improve performance.

1.1.2. Geometry Processing

During this stage, the vertices defined in the geometry stage are processed to prepare them for rendering. There are different spaces in which the vertices are processed:

- Object Space: The original space in which the vertices are defined.

- World Space: The space in which the vertices are transformed to position them in the scene.

- Camera/Eye Space: The space in which the vertices are transformed to position them relative to the camera (the camera is at the origin).

- Clip Space: The space in which the vertices are transformed to prepare them for clipping (removing parts of the geometry that are outside the view frustum).

- Normalized Device Space: The space in which the vertices are transformed to fit within a standardized coordinate system (usually a cube from -1 to 1 in all axes).

- Screen Space: The final space in which the vertices are transformed to fit the actual screen dimensions.

In order to transform the vertices from one space to another, several transformations are applied.

Model Transformation

Before moving again to computer graphics, we should focus on the transformations that we can apply to the vertices.

Scaling It is a linear application that changes the size of an object. It can be uniform (same scaling factor for all axes) or non-uniform (different scaling factors for each axis). An scaling transformation of values (α, β, γ) can be represented in 3D using the matrix (1.1).

$$\begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{pmatrix} \tag{1.1}$$

The inverse of a scaling transformation is also a scaling transformation with the reciprocal of the scaling factors.

Shearing It is a linear application that distorts the shape of an object by shifting its vertices in a specific direction depending on their distance to a reference plane. A shearing transformation of value α in the x direction depending on their distance to the XZ plane can be represented in 3D using the matrix (1.2).

$$\begin{pmatrix} 1 & \alpha & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.2}$$

Rotation It is a linear application that rotates an object around a specific axis by a certain angle. A rotation transformation of angle θ around the z axis can be represented in 3D using the matrix (1.3).

$$\begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.3}$$

The inverse of a rotation transformation is also a rotation transformation with the opposite angle. Mathematically, given that the rotation matrix is orthogonal, its inverse is equal to its transpose.

Translation This last type of transformation is not a linear application, as it does not preserve the origin. These transformations need to be represented using homogeneous coordinates, which add an extra dimension to the vertices. Therefore:

- Points are represented as $(x, y, z, 1)^t$.
- Vectors are represented as $(x, y, z, 0)^t$.

Every transformation that we have allready seen can be represented in homogeneous coordinates by adding an extra row and column to the transformation matrix, with 0s in the new row and column, except for the last element of the

new column, which is 1. For example, a translation transformation of values (α, β, γ) can be represented in 3D using the matrix (1.4).

$$\begin{pmatrix} 1 & 0 & 0 & \alpha \\ 0 & 1 & 0 & \beta \\ 0 & 0 & 1 & \gamma \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{1.4}$$

The model transformation is the transformation that takes the vertices from object space to world space. It is used to position the objects in the scene. It is usually represented as a combination of different transformations, and mathematically it can be represented as a single matrix M_{model} that is the product of the individual transformation matrices (taking the order of transformations into account).

$$P_{\text{world}} = M_{\text{model}} \cdot P_{\text{object}}$$

When a transformation M is applied to a vertex P, the normal vector N of the vertex is also transformed. Someone might think that the normal vector can be transformed using the same transformation M, but this is not always the case. The normal vector should be transformed using the inverse transpose of the transformation matrix M to ensure that it remains perpendicular to the surface after the transformation. Mathematically, the transformed normal vector N' can be calculated as follows:

$$N' = (M^{-1})^t \cdot N$$

Demostración. To understand why the normal vector should be transformed using the inverse transpose of the transformation matrix, we need to consider how transformations affect the geometry of the surface. Let n be the normal vector of a surface at a point P and t be the tangent vector at the same point. For clarity, let's use T_O to denote the transformation applied to the points (the tangent) and T_N to denote the transformation applied to the normal vector. We want to ensure that after applying the transformations, the normal vector remains perpendicular to the tangent vector, which means:

$$(T_N \cdot n) \perp (T_O \cdot t) \iff (T_N \cdot n) \cdot (T_O \cdot t) = 0 \iff (T_N \cdot n)^t \cdot (T_O \cdot t) = 0 \iff n^t \cdot T_N^t \cdot T_O \cdot t = 0$$

Given that n is perpendicular to t, we have $n^t \cdot t = 0$. Therefore, for the transformed normal vector to remain perpendicular to the transformed tangent vector, we need:

$$T_N^t \cdot T_O = I \iff T_N^t = T_O^{-1} \iff T_N = (T_O^{-1})^t$$

□

This should always be taken into account when applying transforming normals, *the normal vector should be transformed using the inverse transpose of the transformation matrix applied to the points.*

View Transformation

The view transformation is the transformation that takes the vertices from world space to camera/eye space. It is used to position the camera in the scene and to define the view direction. In OpenGL, the camera is always positioned at the origin of the coordinate system, looking in the z direction. As the camera can't move, the view transformation is achieved by applying the inverse of the camera's transformation to the vertices.

Let's say that we want to position the camera at a point C in world space, looking at a direction D. We can define the view transformation using the following steps:

1. First of all, as we want to position the camera at point C, we need to apply a translation transformation that moves the world in the opposite direction of C. This can be represented using a translation matrix T that translates by $-C$.

2. Next, we need to align the camera's view direction with the z axis. To do this, we only know the direction D that the camera is looking at.

   a) We can calculate the right vector R of the camera using the cross product of the view direction D and the up vector U' (which is usually defined as $(0, 1, 0)$). This can be represented as $R = D \times U$.

   b) Then, we can calculate the up vector U of the camera using the cross product of the right vector R and the view direction D. This can be represented as $U = R \times D$.

   This way, the matrix $[R, U, D]$ maps this orthogonal basis to the standard basis of the camera space. Therefore, the inverse of this transformation is:

$$[R, U, D]^{-1}$$

Therefore, the view transformation can be represented as the product of the translation and the inverse of the rotation:

$$M_{\text{view}} = [R, U, D]^{-1} \cdot T$$

Lighting and Shading

Before moving to the next stage, lighting and shading should be considered, because they are the processes that determine the color and intensity of the pixels in the final rendered image. Lighting is the process of calculating how light interacts with the surfaces of the objects in the scene, while shading is the process of determining the color of each pixel based on the lighting calculations. The Phong Lighting Model is a widely used model for calculating the lighting in computer graphics. It uses the following notation.

Notación. The notation used in the Phong Lighting Model is shwon in Figure 1.1.

- X is the point on the surface being shaded.

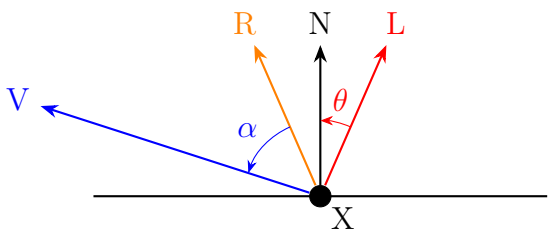

Figura 1.1: Notation used in the Phong Lighting Model.

- N is the normal unitary vector at point X.
- L is the unitary vector from point X to the light source.
- V is the unitary vector from point X to the viewer (camera).
- R is the reflection of the light vector L around the normal vector N, calculated as $R = 2(N \cdot L)N - L$.

  This calculation is valid because $N \cdot L = \cos(\theta)$ gives the projection of L onto N (as the cosine is the adjacent side of the triangle formed by L and N).

The Phong Lighting Model calculates the color of a pixel using three components: ambient, diffuse, and specular.

- Ambient Component: This component represents the constant background light in the scene. It is calculated as:

  $$\text{Ambient}(X) = C_{X,a} \cdot C_{L,a}$$

  where $C_{X,a}$ is the ambient color of the surface at point X and $C_{L,a}$ is the ambient color of the light source.

- Diffuse Component: This component represents the light that is scattered in all directions when it hits a surface. It is calculated as:

  $$\text{Diffuse}(L, X) = C_{X,d} \cdot C_{L,d} \cdot \text{máx}(0, \cos(\theta)) = C_{X,d} \cdot C_{L,d} \cdot \text{máx}(0, N \cdot L)$$

  where $C_{X,d}$ is the diffuse color of the surface at point X, $C_{L,d}$ is the diffuse color of the light source, and θ is the angle between the normal vector N and the light vector L.

- Specular Component: This component represents the light that is reflected in a specific direction when it hits a surface. It is calculated as:

  $$\text{Specular}(L, X, V) = C_{X,s} \cdot C_{L,s} \cdot \text{máx}(0, \cos(\alpha))^n = C_{X,s} \cdot C_{L,s} \cdot \text{máx}(0, R \cdot V)^n$$

  where $C_{X,s}$ is the specular color of the surface at point X, $C_{L,s}$ is the specular color of the light source, α is the angle between the reflection vector R and the view vector V, and n is the shininess coefficient that controls the size of the specular highlight.

The final color of the pixel is then calculated as the sum of these three components:

$$\text{Color}(L, X, V) = \text{Ambient}(X) + \text{Diffuse}(L, X) + \text{Specular}(L, X, V)$$

Perspective Transform

In this stage, the vertices are transformed from camera/eye space to clip space. The next definition is important.

Definición 1.1 (Frustrum)**.** The view frustum is a geometric shape that represents the volume of space that is visible to the camera. It can be defined in several ways (using the field of view, just the planes...). Two important planes are:

- Near Plane: The plane that is closest to the camera. Objects that are closer than this plane will not be rendered.
- Far Plane: The plane that is farthest from the camera. Objects that are farther than this plane will not be rendered.

Two type of frustrums can be defined:

- Perspective Frustum: A frustum that represents a perspective projection, where objects that are farther from the camera appear smaller. This type of frustum looks like a truncated pyramid, with the near plane being smaller than the far plane.
- Orthographic Frustum: A frustum that represents an orthographic projection, where objects that are farther from the camera appear the same size as objects that are closer to the camera. This type of frustum looks like a rectangular box, with the near and far planes being the same size. This type of projection is often used for 2D rendering or for technical drawings where accurate measurements are important. It represents that the viewer is at the infinity.

The perspective transformation is then used to transform the vertices from camera/eye space to clip space. It is represented as a matrix that prepares the vertices for perspective division. The exact form of the perspective transformation matrix depends on the parameters of the view frustum (how was it given), but it generally includes terms that account for the field of view, aspect ratio, and the near and far planes. We will simply consider the $M_{\text{projection}}$ matrix as the one that transforms the vertices from camera/eye space to clip space, without going into the details of its construction. Therefore, what the programmer should implement in OpenGL is:

$$P_{\text{clip}} = M_{\text{projection}} \cdot M_{\text{view}} \cdot M_{\text{model}} \cdot P_{\text{object}}$$

Clipping

The clipping stage removes any vertices that are outside the view frustum. Here, the vertices are transformed from clip space to normalized device space by performing perspective division. Given that the vertices are in homogeneous coordinates and we need the last w coordinate to be 1, the other coordinates are divided by w to achieve this. After perspective division, and given that the correct perspective transformation was applied, the vertices will be in normalized device space, and only the vertices that are within the range of $[-1, 1]$ in all axes will be kept for further processing. The vertices that are outside this range will be clipped, meaning that they will be discarded and not rendered.

Viewport Transformation

In this last stage, the vertices are transformed from normalized device space to screen space. This transformation maps the normalized device coordinates to the actual pixel coordinates on the screen. The viewport transformation is defined by the viewport parameters, which specify the position and size of the viewport on the screen. The viewport transformation can be represented as a matrix that scales and translates the vertices to fit within the specified viewport. After this transformation, the vertices are ready to be rasterized and rendered on the screen.

1.1.3. Rasterization

The rasterization stage is the process of converting the vertices and primitives into pixels on the screen. During this stage, the graphics pipeline determines which pixels on the screen correspond to each primitive, and a fragment is generated for each of these pixels. The data for each vertex (color, texture coordinates, etc.) is interpolated across the primitive to determine the attributes of each fragment.

Data Interpolation

Given $n+1$ points $\overrightarrow{x}_i \in \mathbb{R}^k$ with their corresponding values $f_i \in \mathbb{R}$, $i \in \{0, \ldots, n\}$, interpolating is finding a function $f : \mathbb{R}^k \to \mathbb{R}$ such that $f(\overrightarrow{x}_i) = f_i$ for all $i \in \{0, \ldots, n\}$. When achieved, the function f can be used to estimate the value of f at any point $\overrightarrow{x} \in \mathbb{R}^k$ by evaluating $f(\overrightarrow{x})$.

Linear interpolation is a simple method of interpolation that only considers the linear functions f, which can be represented as $f(\overrightarrow{x}) = \overrightarrow{a} \cdot \overrightarrow{x} + b$ for some $\overrightarrow{a} \in \mathbb{R}^k$ and $b \in \mathbb{R}$. Given that we have $n+1$ constraints (one for each point) and $k+1$ unknowns (the components of $\overrightarrow{a}$ and b), we can solve for the coefficients of the linear function f if $n+1 \leqslant k+1$. In our case, we are interested in interpolating the attributes of the vertices across the primitives, which means that we have 3 vertices (for triangles) and two dimensions (the screen coordinates), so $k = 2$. Therefore, our aim is to find $a, b, c \in \mathbb{R}$ such that $f(x, y) = ax + by + c$ and $f(\overrightarrow{x}_i) = f_i$ for $i \in \{0, 1, 2\}$. This can be achieved by solving the system of equations of Equation (1.5).

$$\begin{cases} ax_0 + by_0 + c = f_0 \\ ax_1 + by_1 + c = f_1 \\ ax_2 + by_2 + c = f_2 \end{cases} \implies \begin{pmatrix} x_0 & y_0 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} f_0 \\ f_1 \\ f_2 \end{pmatrix} \tag{1.5}$$

However, this does not take the perspective into account. To achieve perspective-correct interpolation, we need to consider the depth of each vertex.

Shading

Shading is the process of determining the color of each pixel based on the lighting calculations. During rasterization, the attributes of the vertices are interpolated across the primitive to determine the attributes of each fragment. There are different ways to apply the Phong Lighting Mode, depending on what is interpolated and when is the color calculated:

- Flat Shading: The color is calculated for each primitive (e.g., triangle) using a single value of each attribute for the whole primitive. This results in a faceted appearance, as each primitive has a single color.
- Gouraud Shading: The color is calculated at each vertex , and then interpolated across the primitive. This results in a smoother appearance than flat shading, but it can produce artifacts such as Mach bands.
- Phong Shading: The normal vector is interpolated across the primitive, and the color is calculated for each fragment using the interpolated normal vector. This results in a smoother appearance than Gouraud shading and can produce more accurate lighting effects.

1.1.4. Fragment Processing

The fragment processing stage is almost the final stage of the graphics pipeline, where the fragments generated during rasterization are processed to determine their final color and whether they should be rendered on the screen. During this stage, several operations can be performed on the fragments, such as depth testing, stencil testing, blending, and more. The specific operations performed during fragment processing depend on the settings and configurations of the graphics pipeline.

Textures

A texture map is a discretely sampled multi-dimensional function containing material properties, like color or normals. Texture mapping is the process of applying a texture map to a 3D model to add detail and realism to the rendered image. During fragment processing, the texture coordinates of each fragment are used to sample the corresponding attribute from the texture map, which can then be used. Some concepts about textures are:

- Wrapping: Texture coordinates usually are planned to be in the range of $[0, 1]$, but sometimes they can be outside this range. Wrapping is the process of determining how to handle texture coordinates that are outside the $[0, 1]$ range. Common wrapping modes include:
  - `GL_REPEAT`: The texture is repeated when the texture coordinates are outside the $[0, 1]$ range.
  - `GL_MIRRORED_REPEAT`: The texture is repeated and mirrored when the texture coordinates are outside the $[0, 1]$ range.
  - `GL_CLAMP_TO_EDGE`: The texture coordinates are clamped to the edge of the texture when they are outside the $[0, 1]$ range. This means that the texture will be stretched to fill the area outside the $[0, 1]$ range.

  One mode should be selected for each texture coordinate (e.g., s and t).
- Filtering: Usually, the texture maps are of a different resolution than the rendered image, which means that the texture needs to be filtered to determine the color of each pixel. There are two types of filtering:

- `Minification`: The process of determining the color of a fragment when the texels (texture pixels) are smaller than the screen pixels.
- `Magnification`: The process of determining the color of a fragment when the texels are larger than the screen pixels.

Common filtering modes include:

- `GL_NEAREST`: The color of the fragment is determined by the color of the nearest texel. In minification, this can lead to aliasing artifacts, while in magnification, it can lead to a blocky appearance.
- `GL_LINEAR`: The color of the fragment is determined by the bilinear interpolation of the colors of the 4 nearest texels. In minification it does not make a big difference (considering 4 texels when the pixel may be covered by 100 texels does not improve the quality), but in magnification it can improve the appearance of the texture, making it appear smoother.

  In order to achieve bilinear interpolation, it is simply interpolated in both dimensions.

- MIP Mapping: MIP mapping is a technique used to improve the quality of textures when they are minificated. It involves creating multiple levels of detail for a texture, where each level is a downscaled version of the original texture. During rendering, the appropriate level of detail is selected based on the distance and angle of the fragment being rendered. This can help to reduce aliasing artifacts and improve the overall appearance of the texture, as no minification is applied when the appropriate level of detail is selected.

Fragment Tests

After the fragment processing stage, several tests can be performed on the fragments to determine whether they should be rendered on the screen or discarded. Before performing these tests, the content of each pixel should be explained. Each pixel (not fragment) on the screen has a buffer that stores:

- Color: The color of the pixel, usually represented as RGBA (red, green, blue, alpha).
- Depth: The depth value of the current pixel, which is the depth value of the fragment that is currently rendered at that pixel (the closest fragment to the camera).
- Stencil: The stencil value of the pixel, which is used for stencil testing. The programmer can define the stencil value for each pixel and store it in the stencil buffer.

Once this has been explained, some common fragment tests include:

- Alpha Test: This test discards fragments based on their alpha value. It consists of a comparison between the alpha value of the fragment and a reference value, using a specified comparison function (e.g., less than, greater than, equal to). If the comparison fails, the fragment is discarded and not rendered on the screen.

- Stencil Test: This test discards fragments based on their stencil value. It compares the stencil value of the pixel where the fragment would be rendered with a reference value, using a specified comparison function. If the comparison fails, the fragment is discarded and not rendered on the screen. The stencil test can be used for various effects, such as creating shadows, outlining objects, or implementing complex masking operations.
- Depth Test: This test discards fragments based on their depth value. It compares the depth value of the fragment with the depth value of the corresponding pixel in the depth buffer. If the fragment is closer to the camera than the existing pixel, it passes the depth test and is rendered on the screen, and the depth buffer is updated with the new depth value. If the fragment is farther from the camera than the existing pixel, it fails the depth test and is discarded.

Blending

Blending is the process of combining the color of a fragment with the color of the corresponding pixel in the framebuffer to produce a final color that is rendered on the screen. Blending is typically (but not only) used to achieve transparency effects, where the color of the fragment is blended with the color of the background to create a semi-transparent appearance. The blending operation is defined by:

$$C_{\text{final}} = C_{\text{source}} \cdot F_{\text{source}} \odot C_{\text{destination}} \cdot F_{\text{destination}}$$

where:

- C_{final} is the final color that is rendered on the screen.
- C_{source} is the color of the fragment being processed.
- $C_{\text{destination}}$ is the color of the corresponding pixel in the framebuffer.
- F_{source} and $F_{\text{destination}}$ are the blending factors for the source and destination colors, respectively. These factors determine how much of each color contributes to the final color. Common blending factors include:
  - `GL_ZERO`: The factor is 0, meaning that the corresponding color does not contribute to the final color.
  - `GL_ONE`: The factor is 1, meaning that the corresponding color fully contributes to the final color.
  - `GL_SRC_ALPHA`: The factor is equal to the alpha value of the source color, meaning that the contribution of the source color is proportional to its transparency.
  - `GL_ONE_MINUS_SRC_ALPHA`: The factor is equal to 1 minus the alpha value of the source color, meaning that the contribution of the destination color is proportional to the transparency of the source color.

  The specific blending factors used can be configured based on the desired effect.
- $\odot$ represents the operator used to combine the source and destination colors. Common operators include addition, subtraction, minimum, and maximum.

1.1.5. Framebuffer

The framebuffer is the final destination for the rendered image. It is a memory buffer that stores the color, depth, and stencil information for each pixel on the screen. After all the fragment processing and tests have been performed, the final color of each pixel is written to the framebuffer, which is then displayed on the screen. The framebuffer can also be used for off-screen rendering, where the rendered image is stored in a texture or a renderbuffer instead of being displayed on the screen. This allows for various effects and techniques, such as post-processing, shadow mapping, and more.

1.2. Programmable Pipeline

The programmable pipeline is a modern graphics pipeline that allows developers to write custom shaders to control the behavior of each stage of the pipeline. This provides greater flexibility and control over the rendering process, allowing for more complex and realistic graphics. The programmable pipeline consists of several stages, including vertex shading, tessellation, geometry shading, fragment shading, and more. Each stage can be programmed using a shading language such as GLSL (OpenGL Shading Language) or HLSL (High-Level Shading Language). The programmable pipeline has largely replaced the fixed-function pipeline in modern graphics hardware, as it allows for more advanced rendering techniques and greater performance.

2. Reflection

During this chapter, we will discuss how can reflections be simulated in computer graphics. We will start with the simplest case of planar reflections, and then we will move to more complex cases of non-planar reflections and refraction.

2.1. Planar Reflection

A planar reflection is a reflection that occurs on a flat surface, such as a mirror or a calm body of water. Let's first approach it from a mathematical perspective, and then we will see how it can be implemented in computer graphics. Let p be a point in a surface with normal vector N, and let q be the point that we want to reflect over the surface, obtaining the reflected point q_m. Let α be the angle between the vector $\overrightarrow{pq}$ and the normal vector N. Our aim is to find the coordinates of q_m given the coordinates of p, q, and N. The situation is illustrated in the Figure 2.1.

Given that the surface is planar, we can q_m as:

$$q_m = q - 2N \cdot \text{distance}(q, \text{plane})$$

In order to find that distance, we have:

$$\left.\begin{aligned} N \cdot \overrightarrow{pq} &= \|N\| \cdot \|\overrightarrow{pq}\| \cdot \cos(\alpha) = 1 \cdot \|\overrightarrow{pq}\| \cdot \cos(\alpha) \\ \cos(\alpha) &= \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{\text{distance}(q, \text{plane})}{\|\overrightarrow{pq}\|} \end{aligned}\right\} \Longrightarrow$$

$$\Longrightarrow \text{distance}(q, \text{plane}) = \cos(\alpha) \cdot \|\overrightarrow{pq}\| = N \cdot \overrightarrow{pq}$$

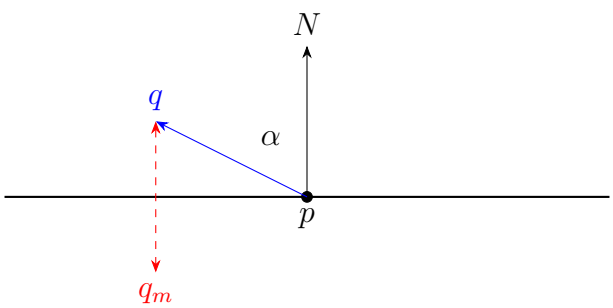

Figura 2.1: Planar reflection.

Therefore, we can express q_m as:

$$\begin{aligned} q_m &= q - 2N(N \cdot \overrightarrow{pq}) \\ &= q - 2N(N \cdot (q - p)) \\ &= q - 2N(N_x(q_x - p_x) + N_y(q_y - p_y) + N_z(q_z - p_z)) \end{aligned}$$

We should take into account that this formula is valid in $3D$. Therefore, the coordinates of q_m can be expressed as:

$$\begin{aligned} q_{m_x} &= q_x - 2N_x(N_x(q_x - p_x) + N_y(q_y - p_y) + N_z(q_z - p_z)) \\ &= (1 - 2N_x^2)q_x + (-2N_xN_y)q_y + (-2N_xN_z)q_z + 2N_x(N_xp_x + N_yp_y + N_zp_z) \\ q_{m_y} &= q_y - 2N_y(N_x(q_x - p_x) + N_y(q_y - p_y) + N_z(q_z - p_z)) \\ &= (-2N_yN_x)q_x + (1 - 2N_y^2)q_y + (-2N_yN_z)q_z + 2N_y(N_xp_x + N_yp_y + N_zp_z) \\ q_{m_z} &= q_z - 2N_z(N_x(q_x - p_x) + N_y(q_y - p_y) + N_z(q_z - p_z)) \\ &= (-2N_zN_x)q_x + (-2N_zN_y)q_y + (1 - 2N_z^2)q_z + 2N_z(N_xp_x + N_yp_y + N_zp_z) \end{aligned}$$

Using $k = N \cdot p$ (treating p as a vector), we can express q_m as:

$$\begin{aligned} q_{m_x} &= (1 - 2N_x^2)q_x + (-2N_xN_y)q_y + (-2N_xN_z)q_z + 2N_xk \\ q_{m_y} &= (-2N_yN_x)q_x + (1 - 2N_y^2)q_y + (-2N_yN_z)q_z + 2N_yk \\ q_{m_z} &= (-2N_zN_x)q_x + (-2N_zN_y)q_y + (1 - 2N_z^2)q_z + 2N_zk \end{aligned}$$

Therefore, we have the following matrix representation of the planar reflection:

$$\begin{pmatrix} q_{m_x} \\ q_{m_y} \\ q_{m_z} \\ 1 \end{pmatrix} = \begin{pmatrix} 1 - 2N_x^2 & -2N_xN_y & -2N_xN_z & 2N_xk \\ -2N_yN_x & 1 - 2N_y^2 & -2N_yN_z & 2N_yk \\ -2N_zN_x & -2N_zN_y & 1 - 2N_z^2 & 2N_zk \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \\ q_z \\ 1 \end{pmatrix}$$

This matrix can be used to perform the planar reflection in computer graphics. We can apply this transformation to the vertices of the objects that we want to reflect, and then we can render the reflected objects as usual.

One first approach to implement planar reflections in computer graphics would be:

1. Render the reflected objects using the planar reflection transformation.
2. Disable color buffer writing, so that the mirror surface is not rendered (just the depth buffer is updated).
3. Render the mirror surface.
4. Enable color buffer writing.
5. Render the original objects.

However, this approach has some issues, such as the fact that the reflected objects are not clipped by the mirror surface. The stencil buffer can be used to solve this issue, by marking the pixels that belong to the mirror surface and then using this information to clip the reflected objects. In addition, a clip plane is used to avoid rendering the parts of the reflected objects that are in front of the mirror surface.

2.1.1. Recursive Reflections

Recursive reflections occur when an object is reflected in a mirror, and then this reflection is reflected again in another mirror, and so on. This can lead to an infinite number of reflections, which can be computationally expensive to render. One way to handle recursive reflections is to limit the number of reflections that are rendered, which will reduce the computational cost while still providing a visually appealing result. However, this approach can lead to artifacts, such as the fact that the reflections will not be accurate after a certain number of reflections. This leads to the need of finding alternative solutions to handle recursive reflections which can provide a more accurate result while still being computationally efficient.

Alternative Solution

An alternative solution to handle planar reflections is to use a technique called "render-to-texture". This technique involves reflecting the camera across the mirror surface (not only the position, but also the direction), and then rendering the reflected scene from there. This rendered image can then be used as a texture to apply to the mirror surface, which will give the illusion of a reflection. This technique is more efficient than rendering every reflection recursively, as it only requires rendering the scene once.

2.2. Non-Planar Reflection

Non-planar reflections occur when the reflecting surface is not flat, such as a curved mirror or a body of water with waves. In this case, the reflection is more complex to calculate, as the normal vector at each point of the surface can vary. One way to handle non-planar reflections is to use a technique called "environment mapping", which uses "cube maps" to store the reflection information for each direction.

A cube map is a texture that consists of six square images, each representing the reflection in a different direction (positive and negative x, y, and z). In order to generate the cube map, the camera is placed at the center of the reflecting surface, and the scene is rendered six times, each time with a different view direction corresponding to one of the faces of the cube map. Once the cube map is generated, it can be used to sample the reflection information for each pixel on the reflecting surface. The cube map has two important properties:

- It is view-independent, meaning that the reflection information is the same regardless of the position of the viewer. It only has to be updated when the reflecting surface moves or the scene changes.
- It is an approximation of the reflection, as it is only phisically perfect at the center of the reflecting surface.

The texture coordinates for sampling the cube map are however easy to calculate.

1. First of all, the reflexion vector is calculated. Let p be the point on the reflecting surface whose reflection we want to calculate, and let N be the normal vector

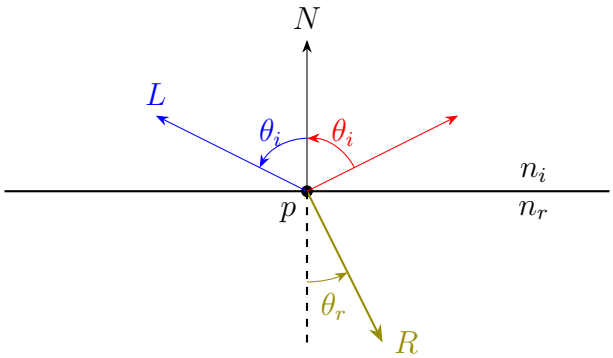

Figura 2.2: Refraction of light.

at point p. Let V be the vector from point p to the viewer (camera). The reflection vector R can be calculated as:

$$R = 2(N \cdot V)N - V$$

2. Once we have the reflection vector R, the largest absolute component of R represents the direction of the reflection. For example, if $|R_x|$ is the largest component, then the reflection is in the x direction. Then, the sign of that component determines whether the reflection is in the positive or negative direction. For example, if R_x is positive, then the reflection is in the positive x direction, and if R_x is negative, then the reflection is in the negative x direction. This information allows us to determine which face of the cube map we need to sample from.

3. Finally, the other two components of the reflection vector R are divided by the largest component to obtain the 2D texture coordinates for sampling the cube map in a range of $[-1, 1]$.

2.3. Refraction

The refraction of light occurs when light passes from one medium to another, such as from air to water. This causes the light to bend, which can create interesting visual effects. The amount of bending depends on the angle of incidence and the refractive indices of the two media. The refractive index is a measure of how much a material slows down light compared to vacuum. The relationship between the angle of incidence, the angle of refraction, and the refractive indices is given by Snell's Law:

$$n_i \sin(\theta_i) = n_r \sin(\theta_r)$$

where n_i and n_r are the refractive indices of the first and second media, respectively, and θ_i and θ_r are the angles of incidence and refraction, respectively. The situation is illustrated in the Figure 2.2.

2.4. Fresnel Effects

The Fresnel effects describes the attenuation of reflected light depending on the angle of incidence, the frequency of the light, and the material properties. The equations for calculating the Fresnel effects are difficult to solve, but they are usually approximated.

3. Transparency

During this chapter, we will discuss the concept of transparency in the context of Image Synthesis. Transparency is a crucial aspect of computer graphics that allows for the creation of images with varying levels of opacity. It enables the blending of different layers and the creation of complex visual effects.

Usually, we render with the depth test enabled, resulting in a final image where only the closest fragments are visible. In order to allow transparency, this test is disabled and the fragments are blended together. The blending process is controlled by the blending function and the blending factors, as explained in Section 1.1.4.

- The *Blending Function* is the addition.
- The source blending factor is the alpha value of the fragment, and the destination blending factor is one minus the alpha value of the fragment. This allows for the creation of semi-transparent effects, where the color of the fragment is blended with the color of the background based on the alpha value of the fragment in front of it.

Therefore, the blending operation can be expressed as in Equation 3.1, and the code needed is shown in Listing 1.

$$C_{\text{final}} = \alpha_{\text{source}} \cdot C_{\text{source}} + (1 - \alpha_{\text{source}}) \cdot C_{\text{destination}} \tag{3.1}$$

However, this has a big problem: *blending is not commutative*. This means that the order in which the fragments are blended together matters, and it can lead to incorrect results if the fragments are not blended in the correct order. To solve this problem, we need to sort the fragments based on their depth values before blending them together. Therefore, the general approach is:

1. Render all opaque objects first, with the depth test enabled and blending disabled.

```
glEnable(GL_BLEND);
glBlendFunc(GL_SRC_ALPHA, GL_ONE_MINUS_SRC_ALPHA);
glBlendEquation(GL_FUNC_ADD);
```

Código fuente 1: Code for enabling blending with the specified blending function and factors.

2. Disable the test buffer updates, but enable the depth test and blending.

3. Render all transparent objects, sorted from farthest to nearest, with blending enabled.

However, this approach has a big problem: the sorting step can be computationally expensive (or even not possible) in scenes with a large number of transparent objects or with complex geometry. For instance, in a situation where three transparent objects are intersecting each other, it may not be possible to determine a clear sorting order for the fragments. In such cases, alternative techniques such as depth peeling can be used to achieve correct blending results without the need for sorting.

3.1. Depth Peeling

Depth peeling is a technique used to achieve correct blending results for transparent objects without the need for sorting. It works by rendering the scene multiple times, each time peeling away the closest layer of fragments. This allows for the correct blending of fragments based on their depth values, even in cases where sorting is not possible. The DP algorithm uses two depth buffers: one for storing the depth values of the closest fragments (reference), and another for storing the depth values of the next closest fragments (the one used in each layer). The algorithm can be summarized as follows:

1. In the first pass, render the scene with the depth test enabled and blending disabled, and store the depth values of the closest fragments in the reference depth buffer.

   This first layer of fragments is stored.

2. In the following passes, the depth buffer from the previous pass is used to peel away fragments with depth values less than or equal to the stored depth values. This allows to ignore the fragments that have already been rendered in the previous passes.

   All other fragments are rendered using the working depth buffer, and the depth values of the next closest fragments are stored in the reference depth buffer.

   Each layer of fragments is also stored.

This process is repeated until all layers of fragments have been rendered and blended together. The final result is a correctly blended image of the transparent objects, without the need for sorting. However, obtaining the Depth Complexity (number of layers) is not trivial.

3.1.1. Depth Complexity

There are several techniques to obtain the depth complexity of a scene, which is the number of layers that need to be rendered in the depth peeling algorithm. Some of these techniques include:

- The "logic" approach. Rendering the scene with the depth test disabled and incrementing the stencil buffer for each fragment. The maximum value in the stencil buffer will give the depth complexity of the scene.

  In order to make it more efficient, the maximum value in the stencil buffer can be obtained using a reduction operation, which is a parallel algorithm that obtains the maximum value in a buffer by recursively dividing the buffer into smaller sub-buffers and comparing the values in each sub-buffer to find the maximum value.

- Using *Occlusion Queries*. This technique involves using occlusion queries to ask the GPU how many fragments were rasterized into pixels in the last `draw` call.

  We just use the algorithm described in the previous section, but each time we ask the GPU if any fragments were rasterized in the last pass. If no fragments were rasterized, we can stop the algorithm, as we have already rendered all the layers of fragments in the scene.

3.1.2. Abuse of the MSAA

An important drawback of the depth peeling algorithm is that it requires multiple rendering passes, which can be computationally expensive. However, the MSAA can be abused to achieve a the same effect requiring much less rendering passes.

Multisample Anti-Aliasing (MSAA) is a technique used to improve the quality of rendered images by reducing aliasing artifacts. It works by sampling multiple points within each pixel and averaging the results to produce a smoother final image. However, MSAA can also be used to extract multiple layers of fragments in a single rendering pass. Instead of saving multiple samples per pixel, we can save multiple layers of fragments in the samples.

The stencil buffer is used to control which layer of fragments is being stored in each sample. We set the stencil buffer associated with the sample 0 to 2, the stencil buffer associated with the sample 1 to 3, and so on. Then, for each layer of fragments, it is written to the sample whose stencil buffer value is 2, and then the stencil buffer value is decremented by 1 for every sample.

The initial stencil buffer values start from 2 in order to control that no more layers are stored than the number of samples available. When all the samples have been used, the stencil buffer values will 0 for all the examples but the last one, which will have a stencil buffer value of 1. If all the values are 0, it means that an overflow has occurred.

Lastly, if more layers are needed than the number of samples available, multiple rendering passes are required. The initial stencil buffer values can be set to $kn + 2$, where k is the number of samples available and n is the number of rendering passes already performed. This way, we can skip the first kn layers of fragments, which have already been rendered in the previous passes, and start rendering the next layers

of fragments directly in the samples. This is limited by the maximum value of the stencil buffer, which is 255 in most implementations, so the maximum number of layers that can be rendered using this technique is 255.

4. Shadows

Shadows are an essential aspect of computer graphics that contribute to the realism and depth of a scene. They provide visual cues about the spatial relationships between objects and the light sources in the scene. In this chapter, we will discuss various techniques for generating shadows in computer graphics.

4.1. Projective Shadows

This technique involves projecting the geometry of the shadow-casting object onto a plane that represents the surface on which the shadow will be cast. Let's fist approach this technique from a mathematical perspective. Consider a point light source located at position l, a point q whose shadow we want to calculate, and a plane defined by a point p and a normal vector N. This situation is illustrated in Figure 4.1. The goal is to find the point q_s on the plane where the shadow of point q will be cast.

Given the normal vector $N = (A, B, C)$ of the plane and the point p on the plane, we can derive the equation of the plane as follows:

$$Ax + By + Cz + D = 0, \qquad D = -(Ap_x + Bp_y + Cp_z).$$

The shadow point q_s can be expressed, for certain $t \in \mathbb{R}$, as:

$$q_s = q + t \cdot (\overrightarrow{lq}) = q + t \cdot (q - l)$$

One approach could be to substitute q_s into the plane equation and solve for t. However, a more efficient method is to make firstly an assumption about the position of the light source.

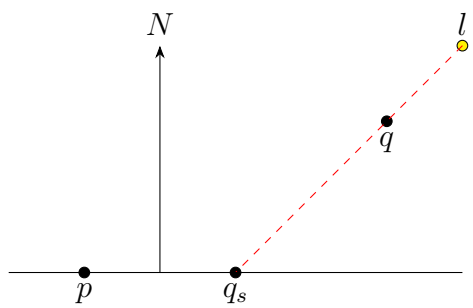

Figura 4.1: Projective shadow calculation.

- Let's firstly suppose that the light source is located at the origin, i.e., $l = (0, 0, 0)$. In this case, the shadow point can be expressed as $q_s = t \cdot q$. Substituting this into the plane equation gives us:

$$A(tq_x) + B(tq_y) + C(tq_z) + D = 0 \implies t = -\frac{D}{Aq_x + Bq_y + Cq_z} = -\frac{D}{\overrightarrow{N} \cdot q}$$

  Therefore, the shadow point can be calculated as:

$$q_s = -\frac{D}{\overrightarrow{N} \cdot q} \cdot q$$

  Given that we work with homogeneous coordinates and that the perspective division will be applied, we have that the shadow point can be calculated as:

$$\begin{pmatrix} -D & 0 & 0 & 0 \\ 0 & -D & 0 & 0 \\ 0 & 0 & -D & 0 \\ A & B & C & 0 \end{pmatrix} \cdot \begin{pmatrix} q_x \\ q_y \\ q_z \\ 1 \end{pmatrix} \equiv \begin{pmatrix} q_{s_x} \\ q_{s_y} \\ q_{s_z} \\ 1 \end{pmatrix}$$

- Now, let's consider the general case where the light source is located at an arbitrary position $l = (l_x, l_y, l_z)$. In this case, let T be the translation matrix for the $\overrightarrow{l}$ vector. Therefore, the shadow point can be calculated as:

$$T \cdot \begin{pmatrix} -D & 0 & 0 & 0 \\ 0 & -D & 0 & 0 \\ 0 & 0 & -D & 0 \\ A & B & C & 0 \end{pmatrix} \cdot T^{-1} \cdot \begin{pmatrix} q_x \\ q_y \\ q_z \\ 1 \end{pmatrix} \equiv \begin{pmatrix} q_{s_x} \\ q_{s_y} \\ q_{s_z} \\ 1 \end{pmatrix}$$

In both cases, we obtain a matrix that can be used to project the shadow of any point q onto the plane defined by p and N. This is then used in the rendering pipeline to create the visual effect of shadows in the scene. After rendering the scene, we can render it again with the shadow projection matrix applied to the geometry of the shadow-casting objects, and the resulting colors will be blended with the background to create the final shadow effect.

Lastly, a bias can be applied to the shadow projection to prevent z-fighting artifacts, which occur when the shadow geometry has the same depth as the surface it is projected onto, causing flickering or other visual artifacts. This bias can be implemented by slightly offsetting the shadow geometry closer to the camera, ensuring that it is rendered in front of the surface and thus avoiding z-fighting.

This technique is relatively simple and efficient, but it has some limitations, because it only works well for flat surfaces and can produce artifacts when the shadow-casting object is close to the surface. Additionally, it does not account for self-shadowing or shadows cast by other objects in the scene.

4.2. Shadow Volumes

Shadow volumes are a more advanced technique for generating shadows that can handle complex scenes with multiple light sources and self-shadowing. The idea is to create a volume that represents the region of space that is in shadow, and then use this volume to determine which parts of the scene are in shadow.

4.2.1. Volumes Generation

The first step in the shadow volume technique is to generate the shadow volume for each possible shadow-casting object in the scene. This is done by extruding the silhouette of the object away from the light source to create a volume that represents the region of space that is in shadow. This is done in three steps:

- Possible silhouette edges are extruded to infinity away from the light source, creating the walls of the shadow volume.
- The back-facing occlder's triangles (with respect to the light source) are extruded to infinity, creating the back cap of the shadow volume.
- The front-facing occlder's triangles (with respect to the light source) are "slightly" extruded away from the light source, creating the front cap of the shadow volume.

This process results in a closed volume that represents the region of space that is in shadow. Once the shadow volumes are generated, they can be used in the rendering pipeline to determine which parts of the scene are in shadow.

4.2.2. Rendering with Shadow Volumes

Here, there are two main approaches to render the scene with shadow volumes: the z-pass and z-fail methods. Both methods involve using the stencil buffer to keep track of which pixels are in shadow.

The z-pass method

First of all, the unlit scene is rendered. Then, for each light source, the following steps are performed:

1. The stencil buffer is cleared to zero, color write and depth write are disabled, but the depth test is enabled. It means that it will only update the stencil buffer.
2. Render the front-facing polygons of the shadow volume (with respect to the camera). For each fragment that *passes* the depth test, the stencil buffer of the corresponding pixel is incremented by one. It represents the fact that the ray from the camera to the pixel has entered the shadow volume.

   The stencil buffer is only updated when the depth test passes, which means that only the fragments that are not ocluded by other real geometry of the scene will affect.

3. Render the back-facing polygons of the shadow volume (with respect to the camera). For each fragment that *passes* the depth test, the stencil buffer of the corresponding pixel is decremented by one. It represents the fact that the ray from the camera to the pixel has exited the shadow volume.

4. Finally, render the lit scene, but only for the pixels where the stencil buffer is equal to zero.

    - If it was greater than zero, it means that the ray from the camera to the pixel is still inside a shadow volume, and thus the pixel is in shadow.
    - If it was less that zero it would mean that the ray from the camera to the pixel has exited more shadow volumes than it has entered, which would mean that the camera is inside a shadow volume. However, this would simply not happen because of overflow issues of the stencil buffer, which is typically implemented as an unsigned integer.

As I mentioned before, the z-pass method can produce artifacts when the camera is inside a shadow volume, because the stencil buffer will not be updated correctly. This happens because the near plane clips the shadow volume, causing the front-facing polygons to be clipped and thus not updating the stencil buffer when they should. To solve this issue, the z-fail method can be used.

The z-fail method

The z-fail method is similar to the z-pass method, but instead of counting from the camera to the real-scene geometry, it counts from the infinity to the real-scene geometry. As it happened with the z-pass method, the unlit scene is rendered first. Then, for each light source, the following steps are performed:

1. The stencil buffer is cleared to zero, color write and depth write are disabled, but the depth test is enabled.

2. Render first the back-facing polygons of the shadow volume (with respect to the camera). For each fragment that *fails* the depth test, the stencil buffer of the corresponding pixel is incremented by one. It represents the fact that the ray, after the real-scene geometry, has exited the shadow volume.

3. Render then the front-facing polygons of the shadow volume (with respect to the camera). For each fragment that *fails* the depth test, the stencil buffer of the corresponding pixel is decremented by one. It represents the fact that the ray, after the real-scene geometry, has entered the shadow volume.

4. Render the lit scene, but only for the pixels where the stencil buffer is equal to zero.

    - If it was greater than zero, it means that the ray, after the real-scene geometry, has exited more shadow volumes than it has entered. It means that the real-scene geometry was in shadow, and thus the pixel is in shadow.

- If it was less that zero it would mean that the ray, after the real-scene geometry, has entered more shadow volumes than it has exited. It would mean that the shadow volumes are not closed (because perspective lets you moving the far plane to infinity), which would be a bug in the shadow volume generation process. In addition, this would not be possible because of overflow issues of the stencil buffer, which is typically implemented as an unsigned integer.

As we can see, the z-fail method solves the problems that the z-pass method had, but this technique is still computationally expensive, because it requires rendering the shadow volume geometry multiple times. In addition, volumes have to be regenerated every frame if the light source or the shadow-casting objects are moving, which can further increase the computational cost.

4.3. Shadow Mapping

Shadow mapping is the most widely used technique for generating shadows in real-time applications, such as video games. The idea is to render the scene from the perspective of the light source to create a depth map, which is then used to determine which parts of the scene are in shadow when rendering from the camera's perspective. The process of shadow mapping can be broken down into two main steps: creating the shadow map and rendering the scene with shadows.

4.3.1. Creating the Shadow Map

The first step in shadow mapping is to create the shadow map. This is done by rendering the scene from the perspective of the light source, using a depth buffer to store the distance from the light source to the closest surface at each pixel. The resulting depth map is called the shadow map. The shadow map is typically stored as a texture, where each texel contains the depth value corresponding to the distance from the light source to the closest surface in that direction. The steps in detail to create the shadow map are as follows:

1. Render the scene from the perspective of the light source.

   First of all, the coordinates of the vertices of the scene are transformed to world space (using the inverse of the matrixes needed). Once in world space, we have to multiply the vertices by the following matrixes:

   - Instead of the usual "view matrix" that transforms the vertices from world space to camera space, we use a "light view matrix" that transforms the vertices from world space to light space.
   - A projection matrix that transforms the vertices from light space also has to be used.

   At this point, the scene is rendered from the perspective of the light source, and the depth values are then stored in the depth buffer.

2. The depth buffer is then saved as a texture, which will be used as the shadow map in the next step.

   Before doing so, and given that the coordinates in the clip space are in the range $[-1, 1]$, we have to transform them to the range $[0, 1]$ to be able to store them in a texture. This can be done by multiplying all of the coordinates by $1/2$ and then adding $1/2$ to all of the coordinates. Once done so, the depth buffer can be saved as a texture, which will be used as the shadow map in the next step.

4.3.2. Rendering the Scene with Shadows

The second step in shadow mapping is to render the scene from the camera's perspective, using the shadow map to determine which parts of the scene are in shadow. When rendering the scene as usual, the following steps are performed for each fragment:

1. The world coordinates of the fragment are transformed to light space using the same light view and projection matrixes that were used to create the shadow map. Also the same transformation to the range $[0, 1]$ is applied to the coordinates. This gives us the corresponding texture coordinates in the shadow map.

2. The depth value stored in the shadow map at the corresponding texture coordinates is retrieved. This depth value represents the distance from the light source to the closest surface in that direction.

3. The depth value of the fragment being rendered is compared to the depth value retrieved from the shadow map. Two things may happen:

   - If the depth value of the fragment is greater than the depth value retrieved from the shadow map, it means that the light does not reach the fragment, and thus the fragment is in shadow. In this case, the fragment is shaded accordingly.
   - If the depth value of the fragment is less than or equal to the depth value retrieved from the shadow map, it means that the light reaches the fragment, and thus the fragment is not in shadow. In this case, the fragment is shaded as usual.

In this case, a bias can be applied to the depth value of the fragment being rendered to prevent shadow acne, which is a common artifact in shadow mapping that occurs when the depth values of the fragments being rendered are very close to the depth values stored in the shadow map, causing weird patterns of shadow to appear on the surfaces. This bias can be implemented by adding a small constant value to the depth value of the fragment being rendered, ensuring that it is slightly further away from the light source than the depth values stored in the shadow map, thus preventing shadow acne.

Resolution of the Shadow Map

One of the main issues with shadow mapping is that the resolution of the shadow map can affect the quality of the shadows. If the shadow map has a low resolution, it can result in blocky or pixelated shadows. The quality is specially significant in shadows closer to the camera, because just one texel covers a larger area of the scene and much more fragments will be affected by the same depth value stored in the shadow map. To mitigate this issue, some techniques can be used, such as:

- Increasing the resolution of the shadow map, which can improve the quality of the shadows, but it also increases the computational cost and memory usage.
- Cascaded shadow maps

  This technique involves using multiple shadow maps with different resolutions for different parts of the scene. For example, a high-resolution shadow map can be used for the area close to the camera, while a lower-resolution shadow map can be used for the area further away from the camera. This allows for better quality shadows in the areas where it is most noticeable, while still keeping the computational cost and memory usage manageable.
- Trapezoidal shadow maps

  This idea is based on the observation that the area of the scene that is visible from the light source is often not a square, but rather a trapezoid. By using a trapezoidal shadow map instead of a square one, we can better fit the shadow map to the visible area of the scene, which can improve the quality of the shadows while still keeping the resolution manageable.

It should be noted that all the shadows we have created are hard shadows, which means that they have a sharp edge. In reality, shadows often have soft edges due to the size of the light source and the distance from the shadow-casting object to the surface on which the shadow is cast. To create soft shadows, additional techniques can be used, but they are beyond the scope of this chapter. Physics Based Rendering (PBR) techniques can also be used to create more realistic shadows by simulating the physical properties of light and materials. This will be covered in more detail in next chapters.

5. Radiance Transfer

During this chapter, we will introduce the two differet types of illumination, local and indirect illumination, to end up developping the global illumination. In order to explain these concepts, we first have to introduce some radiometric quantities. Radiometry is the science of measuring electromagnetic radiation, including visible light. It provides a set of quantities that are used to describe the properties of light and how it interacts with surfaces.

Definición 5.1 (Radiant Energy)**.** Radiant energy is the total amount of energy that is carried by a beam of light.

$$Q \qquad [J]$$

Definición 5.2 (Radiant Flux)**.** Also known as Radiant Power, it is the rate at which radiant energy is transferred or emitted. It is defined as the amount of radiant energy per unit time.

$$\Phi = \frac{dQ}{dt} \qquad [W = {}^{J}/_{s}]$$

Definición 5.3 (Irradiance)**.** Irradiance is the amount of radiant flux that is incident on a surface per unit area.

$$E = \frac{d\Phi}{dA} \qquad [{}^{W}/_{m^2}]$$

For the next measures, the solid angle is needed. It is a measure of the amount of space that an object occupies in the field of view of an observer. It is defined as the area of the projection of the object onto a unit sphere centered at the observer. More generally, if the sphere used for the projection has a radius r, and the surface area of the projection is A, the solid angle Ω is defined as:

$$\Omega = \frac{A}{r^2} \qquad [sr]$$

The unit of solid angle is the steradian (sr) (an adimensional unit), which is defined as the solid angle subtended by an area of r^2 on the surface of a sphere of radius r. It is the three-dimensional equivalent of the two-dimensional radian.

Definición 5.4 (Radiant Intensity)**.** Radiant intensity is the amount of radiant flux that is emitted by a source in a particular direction per unit solid angle.

$$I = \frac{d\Phi}{d\overrightarrow{\omega}} \qquad [{}^{W}/_{sr}]$$

Definición 5.5 (Radiance)**.** Radiance is the amount of radiant flux that is emitted, reflected, transmitted or received by a surface per projected unit area and per unit solid angle.

$$L = \frac{d^2\Phi}{dA \cdot \cos(\theta) \cdot d\overrightarrow{\omega}} \qquad [W/m^2 \cdot sr]$$

where θ is the angle between the normal of the surface and the direction of the incoming light. The term $\cos(\theta)$ is used to account for the fact that the effective area of the surface that is illuminated by the light decreases as the angle of incidence increases.

Radiance is the most used quantity in computer graphics, because it is the one that is conserved[1] along a ray of light. This means that the radiance of a ray of light does not change as it travels through space, unless it interacts with a surface. This property makes it easier to compute the illumination of a scene, because we can trace rays of light from the camera to the scene and compute the radiance at each point of intersection.

Observación. So many measures may be confusing. The basic one is the radiant energy, and then the radiant flux is the rate of change of the radiant energy. All of the other measures are derived from the flux energy connecting it with other quantities:

- Irradiance: Flux per unit area.
- Radiant Intensity: Flux per unit solid angle.
- Radiance: Flux per unit area and per unit solid angle.

5.1. Local Illumination

The local illumination model is the simplest one, and it only takes into account the direct light that hits a surface. It does not consider any light that is reflected or refracted by other surfaces. It is explained in the Equation 5.1.

$$L_O = \rho \cdot L_I \cdot \cos(\theta) = \rho \cdot L_I \cdot N \cdot L \tag{5.1}$$

Where L_O is the outgoing radiance, L_i is the incoming radiance, θ is the angle between the normal of the surface and the direction of the incoming light, N is the normal of the surface and L is the direction of the incoming light. The term ρ is the reflectance of the surface, which is a measure of how much light is reflected by the surface. It is a value between 0 and 1.

5.2. Indirect Illumination

Light is not only emitted by light sources as the local illumination model assumes, but it can also be reflected or refracted by other surfaces. This aspects are the ones addressed by the indirect illumination model, which is more accurate than the local

[1] In fact, it only happens in vacuum.

illumination model, but also more computationally expensive. When an observer looks at a point on a surface, three types of rays should be taken into account to determine the illumination of that point:

- Shadow ray: it is traced from the point of intersection to the light source, and it is used to determine if the point is in shadow or not.
- Reflection ray: if the surface is reflective, a reflection ray is traced in the direction of the reflected light, and it is used to determine which object is visible in the reflection.
- Refraction ray: if the surface is transparent, a refraction ray is traced in the direction of the refracted light, and it is used to determine which object is visible through the transparent surface.

5.3. Global Illumination

The global illumination model unifies the local and indirect illumination models, and it represents how light really is. It is usually called Physically Based Rendering (PBR), because it takes into account all the light that is emitted, reflected, transmitted or received by a surface. It is the most accurate model, but it is also the most computationally expensive. It is based on the Rendering Equation 5.2, which is an integral equation that describes the equilibrium of light in a scene.

$$L_O(x, \omega, \lambda) = L_E(x, \omega, \lambda) + \int_\Omega f_r(x, \omega, \omega', \lambda) \cdot L_I(x, \omega', \lambda) \cdot (\omega' \cdot N)\ d\omega' \tag{5.2}$$

where $L_O(x, \omega, \lambda)$ is the outgoing radiance at point x in direction ω and wavelength λ, L_E is the emitted radiance, f_r is the BRDF, L_I is the incoming radiance, N is the normal of the surface and Ω is the hemisphere of directions above the surface. The integral term represents the contribution of all the incoming light that is reflected by the surface, and it is computed by integrating over all the directions in the hemisphere.

Observación. It should be noted that $\text{máx}(0, \omega' \cdot N)$ is not needed, as the domain of integration is the hemisphere of directions above the surface, which means that $\omega' \cdot N$ is always non-negative. However, if the domain of integration were the entire sphere, then the term $\text{máx}(0, \omega' \cdot N)$ would be needed to ensure that only the directions above the surface contribute to the integral.

5.3.1. Bidirectional Reflectance Distribution Function (BRDF)

The BRDF is a function that describes how light is reflected by a surface. It is defined as the fraction of light from an incoming direction ω' that is reflected in an outgoing direction ω at a point x on the surface, for a given wavelength λ. It is denoted as $f_r(x, \omega, \omega', \lambda)$ and it has the following properties:

- $f_r(x, \omega, \omega', \lambda) \geqslant 0$: the BRDF is always non-negative.

- Helmholtz Reciprocity, which states that the BRDF is symmetric with respect to the incoming and outgoing directions. This means that the light may be inverted.
$$f_r(x, \omega, \omega', \lambda) = f_r(x, \omega', \omega, \lambda)$$

- Energy Conservation, which states that the total amount of light reflected by a surface cannot exceed the amount of light that is incident on it.
$$\int_\Omega f_r(x, \omega, \omega', \lambda) \cdot (\omega' \cdot N)\ d\omega' \leqslant 1 \qquad \forall x, \omega, \lambda$$

5.3.2. Algorithms for Global Illumination

As the user may expect, actually computing the global illumination is a very complex task, because it requires solving the rendering equation, which is an integral equation. There are several algorithms that, taking the rendering equation as a base, try to compute the global illumination in a more efficient way. In the next sections, we will introduce some of the most popular algorithms for global illumination.

Recursive Ray Tracing

Recursive ray tracing is a simple algorithm that traces rays of light from the camera to the scene. For each object that is intersected by a ray, two new rays are traced:

- Reflection ray: if the surface is reflective, a reflection ray is traced in the direction of the reflected light.
- Refraction ray: if the surface is transparent, a refraction ray is traced in the direction of the refracted light.

The algorithm continues tracing rays until:

- A ray does not intersect any object
- A ray intersects a light source
- A ray reaches a maximum recursion depth
- A ray contributes less than a certain threshold to the final image

This algorithm, which may appear that does not produce good results, is actually the basis of many other algorithms for global illumination.

Path Tracing

As recursive ray tracing, path tracing is an algorithm that traces rays of light from the camera to the scene. However, for each object that is intersected by a ray, only one new ray is traced, which is chosen randomly according to the BRDF of the surface. As happened with recursive ray tracing, the algorithm continues tracing rays until:

- A ray does not intersect any object
- A ray intersects a light source
- A ray reaches a maximum recursion depth
- A ray contributes less than a certain threshold to the final image

Path tracing is a more efficient algorithm than recursive ray tracing, and it may produce better results, because it takes into account the BRDF of the surface. However, it is still a very computationally expensive algorithm because it requires tracing a large number of rays to produce a good image.

6. Precomputed Radiance Transfer (PRT)

In the previous chapter, we introduced the Rendering Equation (5.2) but realized that it is almost impossible to solve in real-time. Some approaches to simulate the idea that that equation introduced were discussed, as Recursive Ray Tracing and Path Tracing. However, these methods are computationally expensive and do not actually solve the equation. In this chapter, we will introduce a method that allows us to solve the Rendering Equation in real-time, but with some limitations. This method is called Precomputed Radiance Transfer (PRT). The main idea behind PRT is to precompute as much of the lighting information as possible, so that at runtime we can quickly evaluate the lighting for a given scene.

6.1. Assumptions and Simplifications

To make the problem of solving the Rendering Equation tractable, we need to make some assumptions and simplifications. The three main simplifications we will make are:

- Lambertian reflector

  A lambertian reflector is a surface that reflects light equally in all directions. This means that the BRDF for a Lambertian reflector is constant and does not depend on the incoming or outgoing directions. This simplification allows us to ignore the directional dependence of the BRDF, which makes the problem easier to solve. It is defined by the material's albedo ρ, which is the fraction of incoming light that is reflected by the surface. The BRDF for a Lambertian reflector can be expressed as:

  $$f_r(x, \omega, \omega', \lambda) = \frac{\rho}{\pi}$$

  where the factor of π is included to ensure energy conservation, as the reflected light is distributed uniformly over the hemisphere.

- Only first van Neumann Term

  The Rendering Equation itself hides recursive terms, as the incoming radiance L_i can be expressed is itself the outgoing radiance from another point in the scene. This means that to solve the equation, we would need to evaluate it recursively. Each level in the recursion is called a van Neumann term, and

represents the contribution of light that has bounced a certain number of times in the scene. The first van Neumann term represents the direct lighting, which is the contribution of light that comes directly from the light sources to the point being shaded. By only considering the first van Neumann term, we are ignoring any indirect lighting, which is the contribution of light that has bounced off other surfaces in the scene before reaching the point being shaded.

- Distant Lighting

  Distant lighting is a simplification that assumes that the light sources in the scene are infinitely far away. This means that the incoming light can be treated as parallel rays, and the direction of the incoming light does not change across the scene.

  This simplification, along with the previous one, allows us to simplify the incoming radiance term in the Rendering Equation as:

  $$L_i(x, \omega', \lambda) = L_{\text{env}}(\omega') \cdot V(x, \omega')$$

  where $L_{\text{env}}(\omega')$ is the radiance coming from the environment in the direction ω', and $V(x, \omega')$ is a visibility term represents whether the point x is visible from the direction ω'.

Therefore, with these simplifications, the Rendering Equation can be rewritten as the Equation (6.1):

$$L_o(x, \omega, \lambda) = L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \int_{\Omega} L_{\text{env}}(\omega') \cdot V(x, \omega') \cdot (\omega' \cdot N) \, d\omega' \tag{6.1}$$

The term $V(x, \omega') \cdot (\omega' \cdot N)$ is called the transfer function, and it represents how much of the incoming light from the environment is transferred to the outgoing radiance at the point x in the direction ω.

$$T(x, \omega') = V(x, \omega') \cdot \text{máx}(0, \omega' \cdot N)$$

6.2. Converting the Integral into a Dot Product

The idea that PRT introduces is project the possible integrals onto a basis, so that we can simply convert the integral into a dot product. Let's consider the space of all the possible functions with domain A and range B. Given that this space is infinite-dimensional, every basis will have an infinite number of elements. Chosen a basis $\{b_i\}$, we can express any function $f : A \to B$ as a linear combination of the basis elements:

$$f(x) = \sum_{i=0}^{\infty} c_i b_i(x)$$

where c_i are the coefficients of the linear combination:

$$c_i = \langle f(x), b_i(x) \rangle = \int_A f(x) b_i(x) \, dx$$

For the PRT to really simplify the problem, we need to choose an specific type of basis, called an orthogonal basis.

Definición 6.1 (Orthogonal Basis)**.** A basis $\{b_i\}$ is orthogonal if for any two different elements b_i and b_j, the inner product $\langle b_i, b_j \rangle$ is zero. This means:

$$\langle b_i(x), b_j(x) \rangle = \int_A b_i(x) b_j(x)\ dx = \delta_{ij} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

In order to understand the idea that lies behind PRT, let's consider two functions f and g, and an orthogonal basis $\{b_i\}$. Given that we are only interested in approximating the integral of the product of f and g, only the first N coefficients of the linear combination will be relevant. Let's then assume that the projection of f, g onto the basis $\{b_i\}$ is given by:

$$f(x) \approx \sum_{i=0}^{N} c_i b_i(x) \qquad g(x) \approx \sum_{i=0}^{N} d_i b_i(x)$$

Then, the integral of the product of f and g can be approximated as:

$$\int_A f(x)g(x)\ dx \approx \int_A \left(\sum_{i=0}^{N} c_i b_i(x) \right) \left(\sum_{j=0}^{N} d_j b_j(x) \right)\ dx = \sum_{i=0}^{N} \sum_{j=0}^{N} c_i d_j \int_A b_i(x) b_j(x)\ dx \overset{(*)}{=}$$
$$\overset{(*)}{=} \sum_{i=0}^{N} c_i d_i \int_A b_i(x) b_i(x)\ dx = \sum_{i=0}^{N} c_i d_i$$

where in $(*)$ we used the fact that the basis is orthogonal. Therefore, we have approximated the integral of the product of f and g as a dot product of their coefficients in the ortogonal basis.

6.2.1. Spherical harmonics (SH)

The choice of the basis is crucial for the success of PRT. The most common choice for the basis in PRT is the Spherical Harmonics (SH). Spherical Harmonics are a set of *orthogonal* functions defined on the surface of a unit sphere and with possible complex values. However, and given that we are only using real functions, we will only consider the real spherical harmonics. The fact that they are defined on the surface of a unit sphere makes them ideal for representing functions that depend on directions, using just *two coordinates* (the azimuth θ and the inclination ϕ; the radius is fixed to 1). Spherical Harmonics are a family with two parameters:

- Band: $l \in \mathbb{N}$
- Mode: $m \in \mathbb{Z} \cap [-l, l]$

Therefore, they are usually referred to as $Y_l^m(\theta, \phi)$, where θ is the azimuth and ϕ is the inclination. The number of coefficients needed to represent a function using Spherical Harmonics up to band L is given by:

$$N = \sum_{l=0}^{L} (2l+1) = L + 1 + 2 \cdot \frac{L(L+1)}{2} = L + 1 + L(L+1) = (L+1)^2$$

Observación. It should not be confused how many bands are going to be used with up to which band we are going to use.

- Up to band L: we are going to use all the bands from 0 to L, which means that we are going to use $L + 1$ bands, and therefore $(L + 1)^2$ coefficients.
- L bands: we are going to use the bands from 0 to $L - 1$, which means that we are going to use L bands, and therefore L^2 coefficients.

6.2.2. Monte Carlo Integration

The coefficients of the projection of a function f onto the Spherical Harmonics basis have to be computed using an integral over the unit sphere. To compute these integrals, Monte Carlo Integration is usually used.

Monte Carlo Integration is a numerical method for approximating integrals using random sampling. The idea is to sample points from the domain of the function and use the average value of the function at those points to approximate the integral. The more samples we take, the better the approximation will be.

- Monte Carlo Integration is particularly useful for high-dimensional integrals, where traditional numerical methods may struggle.
- It is an abstraction of the Riemann sum, where instead of using a regular grid of points, we use random samples.

Therefore, to compute the coefficients of the projection of a function f onto the Spherical Harmonics basis, we can use Monte Carlo Integration and just sample the function at random points on the unit sphere.

6.3. PRT Pipeline

The PRT pipeline consists of two main stages: the precomputation stage and the rendering stage. The idea is that as much as possible should be precomputed in the first stage, so that the rendering stage can be as fast as possible.

6.3.1. Precomputation Stage

In order to be able to solve the Rendering Equation in real-time, we need to precompute the following information (assuming that we use L bands for approximating):

1. The values of the L^2 SH in each direction. These are no integrals, but they are needed to project the other functions.
2. The projection of the term $L_{\text{env}}(\omega')$ onto the SH basis, which gives us L^2 coefficients for each color channel (R, G, B). For each of the sample directions for the Monte Carlo Integration, we need:
    - The value of the environment map in that direction, which can be obtained by sampling the environment map.

- The value of the SH basis functions in that direction, which can be obtained from the precomputed values of the SH basis functions.

3. For each vertex x in the scene, the projection of the transfer function $T(x, \omega')$ onto the SH basis, which gives us L^2 coefficients for each vertex. For each of the sample directions for the Monte Carlo Integration, we need to compute the following terms:

   - The visibility term $V(x, \omega')$ can be computed using ray tracing, which is a computationally expensive process. However, given that this is done in the precomputation stage, it is not a problem. Depending how far the ray collides, this term will have a higher or lower value.
   - The term $(\omega' \cdot N)$ can be computed using the normal of the vertex and the direction of the incoming light.
   - The value of the SH basis functions in that direction, which can be obtained from the precomputed values of the SH basis functions.

This way, we have precomputed all the information needed to evaluate the Rendering Equation in real-time. The precomputation stage can be computationally expensive, especially for the third step, where we need to compute the visibility term for each vertex and each sample direction. However, this is done offline, so it does not affect the real-time performance of the rendering stage. In this point, we have:

$$L_{\text{env}}(\omega') \approx \sum_{i=0}^{(L-1)^2} l_i b_i(\omega') \qquad T(x, \omega') \approx \sum_{i=0}^{(L-1)^2} t_{x,i} b_i(\omega')$$

6.3.2. Rendering Stage

In the rendering stage, we can now evaluate the Rendering Equation in real-time using the precomputed information. Given a point x and a direction ω, we can compute the outgoing radiance as:

$$\begin{aligned}
L_o(x, \omega, \lambda) &= L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \int_\Omega L_{\text{env}}(\omega') \cdot V(x, \omega') \cdot (\omega' \cdot N)\ d\omega' = \\
&= L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \int_\Omega L_{\text{env}}(\omega') \cdot T(x, \omega')\ d\omega' \approx \\
&\approx L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \int_\Omega \left(\sum_{i=0}^{(L-1)^2} l_i b_i(\omega') \right) \left(\sum_{j=0}^{(L-1)^2} t_{x,j} b_j(\omega') \right)\ d\omega' \overset{(*)}{=} \\
&\overset{(*)}{=} L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \sum_{i=0}^{(L-1)^2} l_i t_{x,i} \int_\Omega b_i(\omega') b_i(\omega')\ d\omega' = \\
&= L_e(x, \omega, \lambda) + \frac{\rho}{\pi} \sum_{i=0}^{(L-1)^2} l_i t_{x,i}
\end{aligned}$$

where in $(*)$ we used the fact that the basis is orthogonal. Therefore, we have approximated the integral of the product of L_{env} and T as a dot product of their

coefficients in the SH basis. This allows us to evaluate the Rendering Equation in real-time, as we only need to compute a dot product of two vectors of size L^2 for each color channel.

One important thing to note is that the SH are rotationally invariant. That means that if we rotate the scene, the coefficients of the projection of the functions onto the SH basis will change, but the values of the functions themselves will not change. This allows us to, instead of recomputing all the coefficients when the scene is rotated, we can just rotate the coefficients of the projection of the functions onto the SH basis. This is done using a rotation matrix. Therefore, we can achieve real-time rendering of the scene even when the scene is rotated, without having to recompute all the coefficients. However, PRT can't be used when the objects in the scene are moving, as the visibility term will change, and therefore we would need to recompute all the coefficients for each vertex, which is not feasible in real-time. Therefore, PRT is only suitable for *static scenes*, where the objects are not moving.